SeeWandel Climate

Interreg
Alpenrhein-Bodensee-Hochrhein

Kofinanziert von der Europäischen Union

Regime Shifts in Plankton Dynamics: What Drives Them and Can We Predict Them?

Dr. Leonardo Capitani
Department of Aquatic Ecology, Eawag · SeeWandel-Climate

www.seewandel.org

Ecomeets @ Eawag, Dübendorf
28 October 2026

Slides and links

Scan to download the slides

SORTEE
Society for Open, Reliable, and Transparent Ecology and Evolutionary Biology

Slides, code and data shared openly
leomarameo7.github.io/science_communication.html

The Pomati lab



Pomati lab
Plankton Ecology, Eawag

SeeWandel-Climate
Lake Constance

Take home message

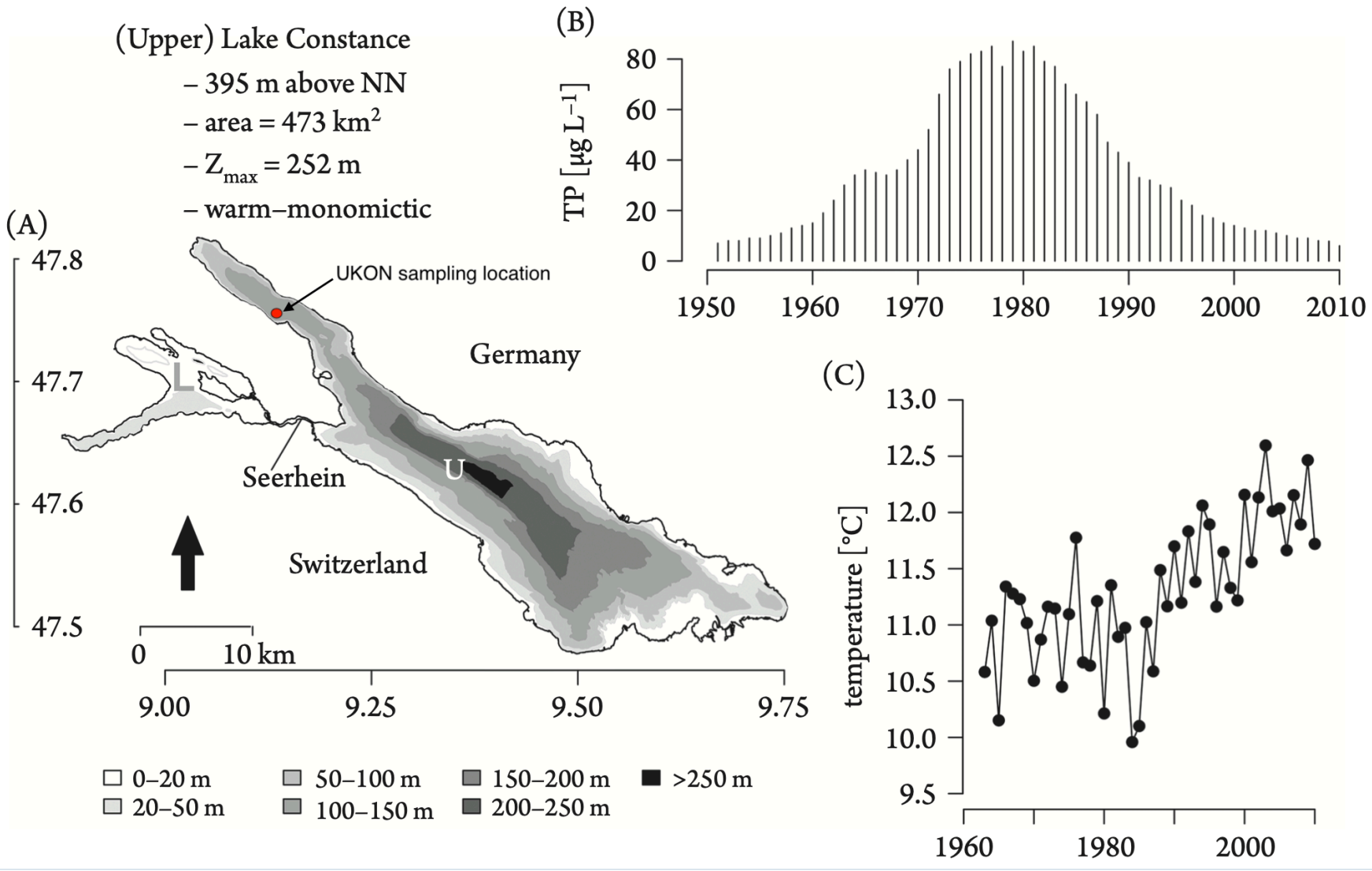
Plankton regime shifts can be detected and partly explained by biotic drivers. Whether they propagate and can be predicted is still open.



Outline

1. What we know: background
2. Research questions
3. Methods: our pipeline
4. Results: detection of regime shifts
5. Explain regime shifts
6. Propagation of regime shifts
7. Conclusions

What we know: background

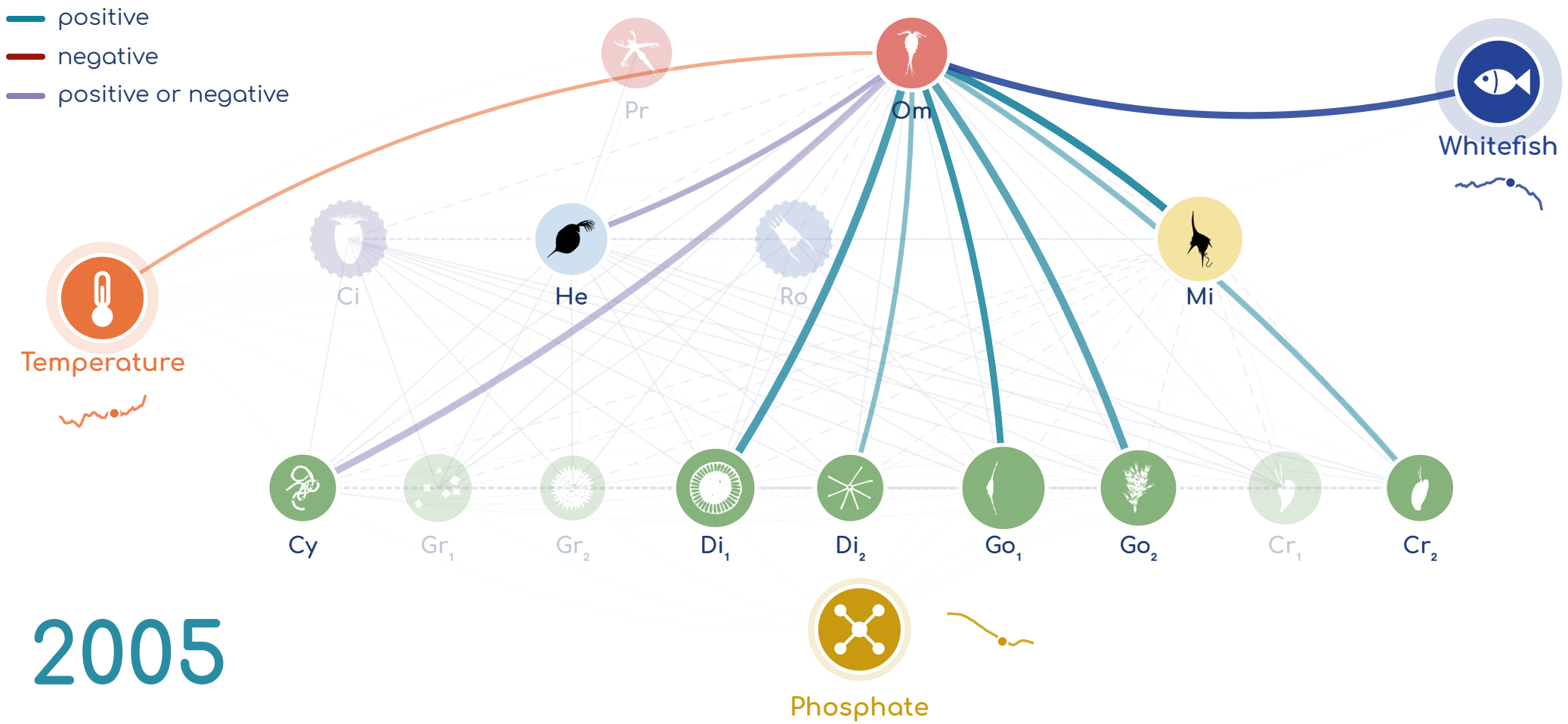


Lake Constance
473 km² · 252 m deep

Phosphate ↑ until 1980, then ↓

Temperature ↑ about 1 °C

What we know: the plankton community



2005

Spotlight: what we expect to drive the omnivores (Om)

Network modified from Merz et al. 2023, Nat. Clim. Change.

Research questions



1 · Detect

Abrupt shifts?



2 · Explain

What drives them?



3 · Propagate

Does a shift spread?



4 · Predict

Can we forecast them? (in progress)

Methods: our pipeline

1

Data

Monthly data 1975–2022

2

Detect

BEAST + two criteria

3

Explain

GP-EDM + hypotheses

4

Propagate

Shift-removal test

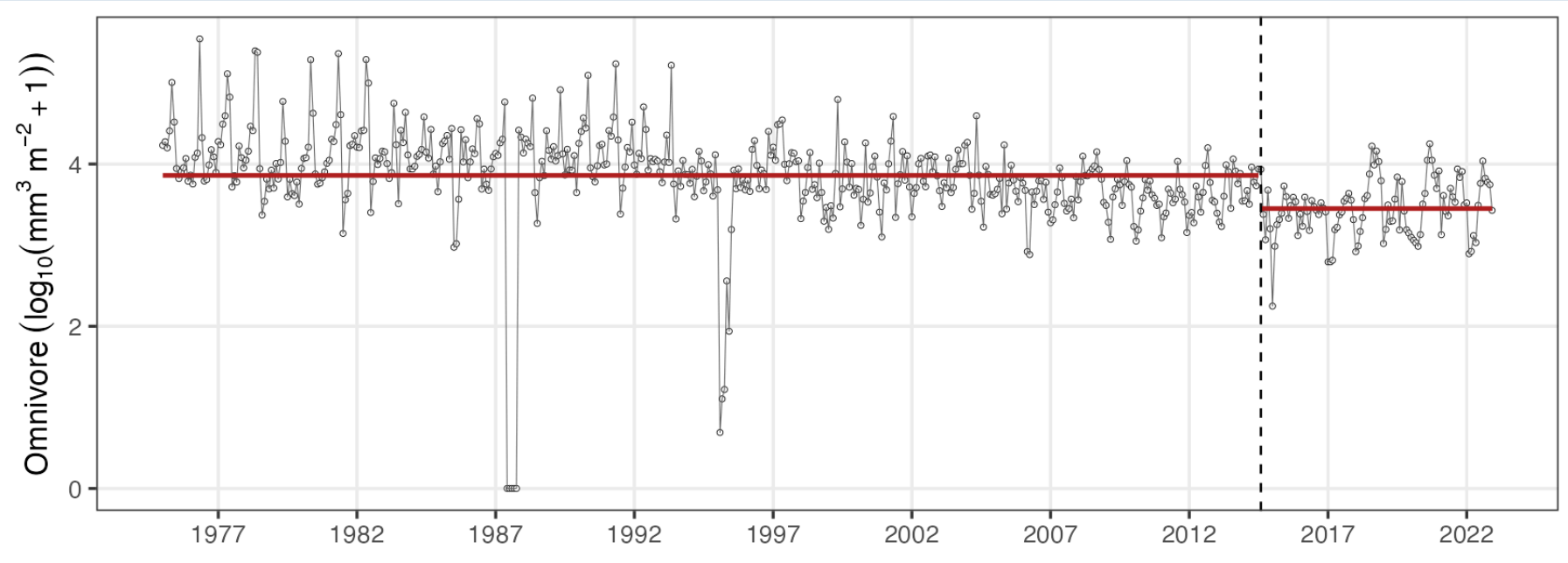
5

Predict

Forward validation

WORK IN PROGRESS

Results: detection of regime shifts



Omnivores, monthly biovolume. Dashed: shift; red: period means.

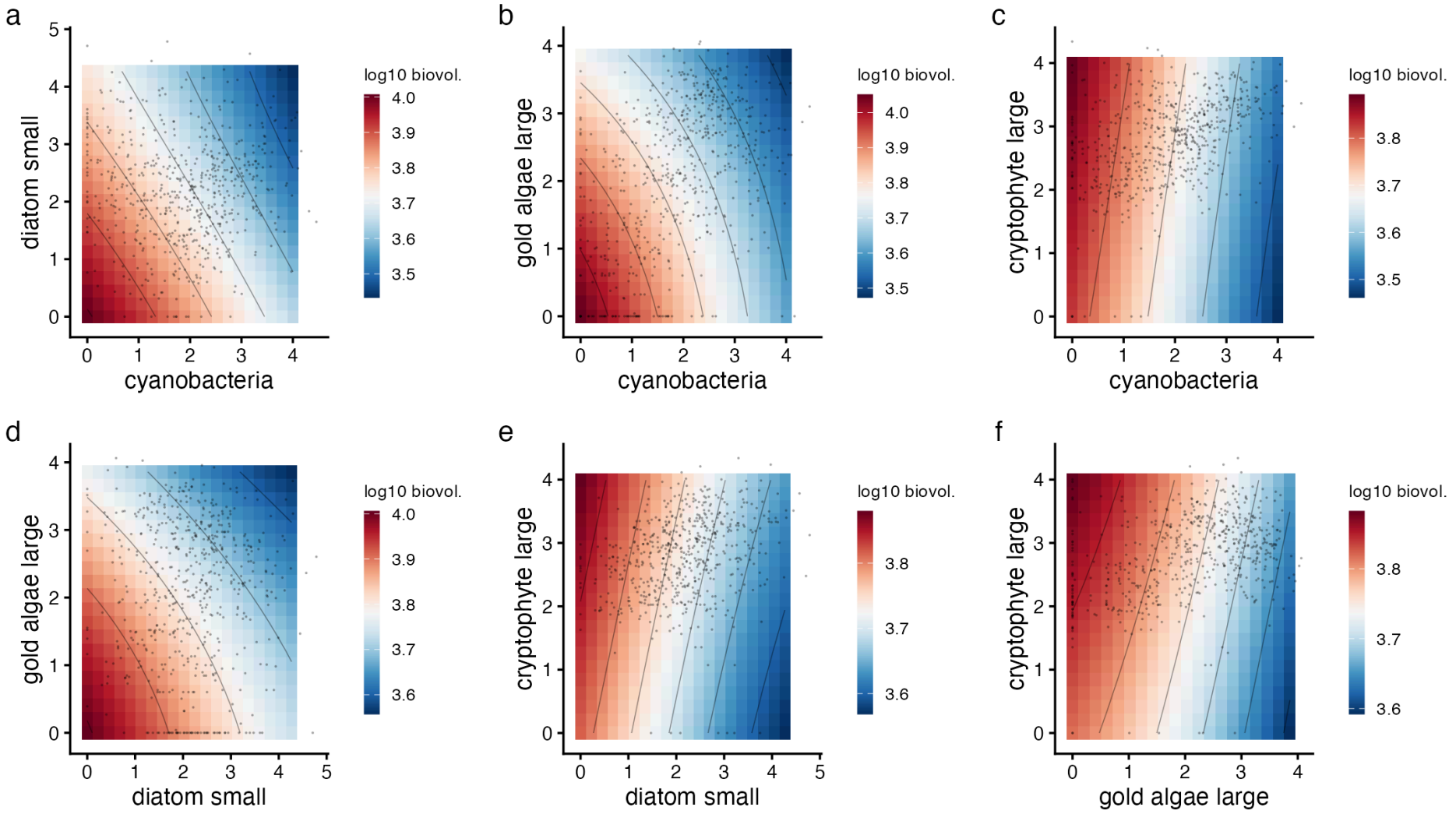


Shift: Aug 2014

abruptness 1.19

The omnivores dropped abruptly in 2014

Explain regime shifts



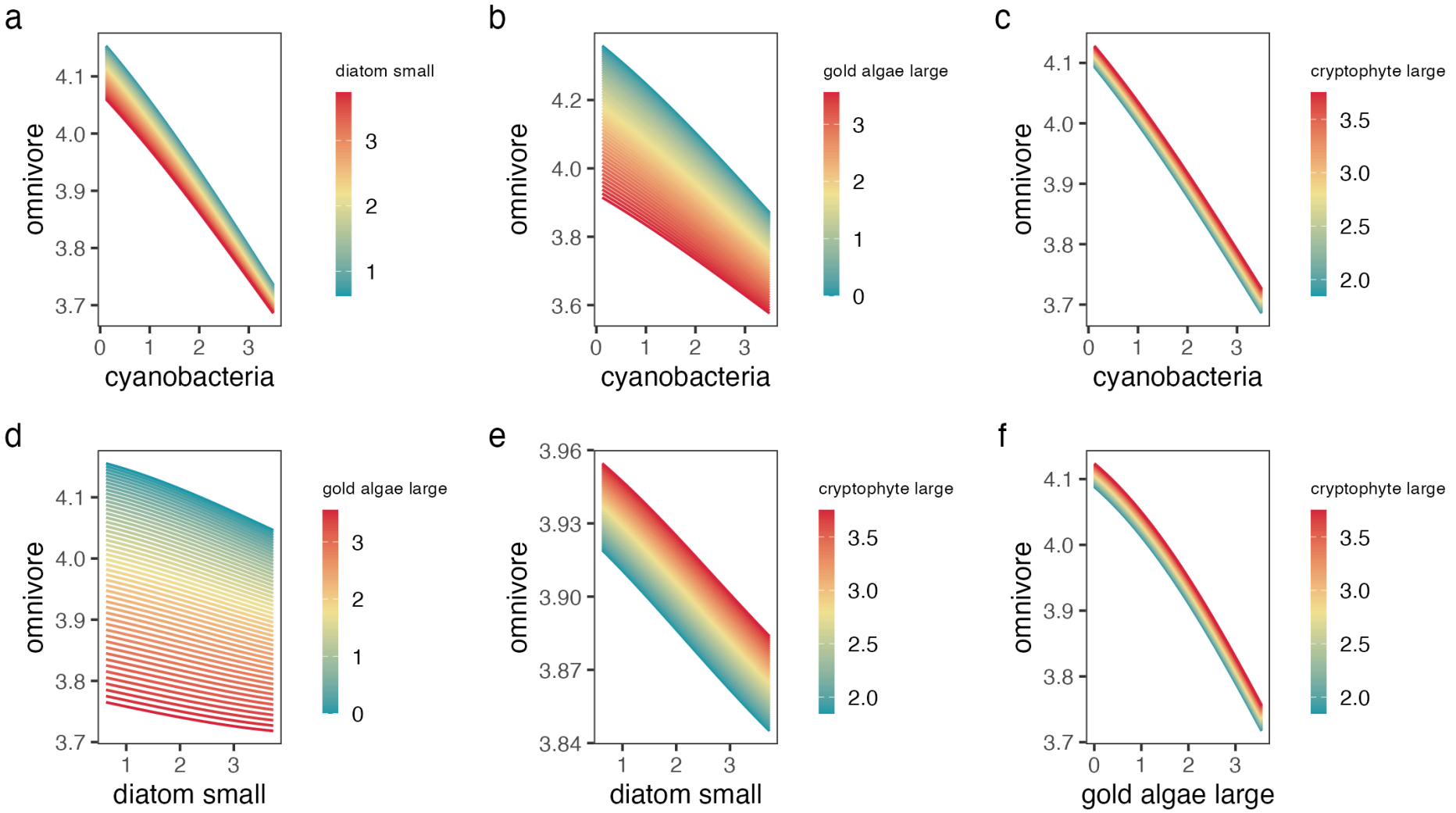


LOO R^2 = **0.81**

Omnivores ↓ with cyanobacteria, small diatoms, large gold algae

Prey hypothesis: **not supported**

Explain regime shifts: partial dependence





x: predictor 1
lines: predictor 2

Cyanobacteria ↓ omnivores

Lines never cross

Propagation of regime shifts

1 · Real predictor



2 candidate pairs

0 testable

Predictor shifted **after**

2 · Shift-free surrogate (× 100)



Gap between real and surrogate prediction = evidence for propagation

No evidence yet: too few shifts

Conclusions



Shifts are detectable



Biotic drivers matter



Propagation, prediction: open

Take home message

Plankton regime shifts can be detected and partly explained by biotic drivers. Whether they propagate and can be predicted is still open.



Acknowledgments

Interreg
Alpenrhein-Bodensee-Hochrhein

Kofinanziert von der Europäischen Union

igkb
Internationale Gewässerschutzkommission für den Bodensee

IBKF

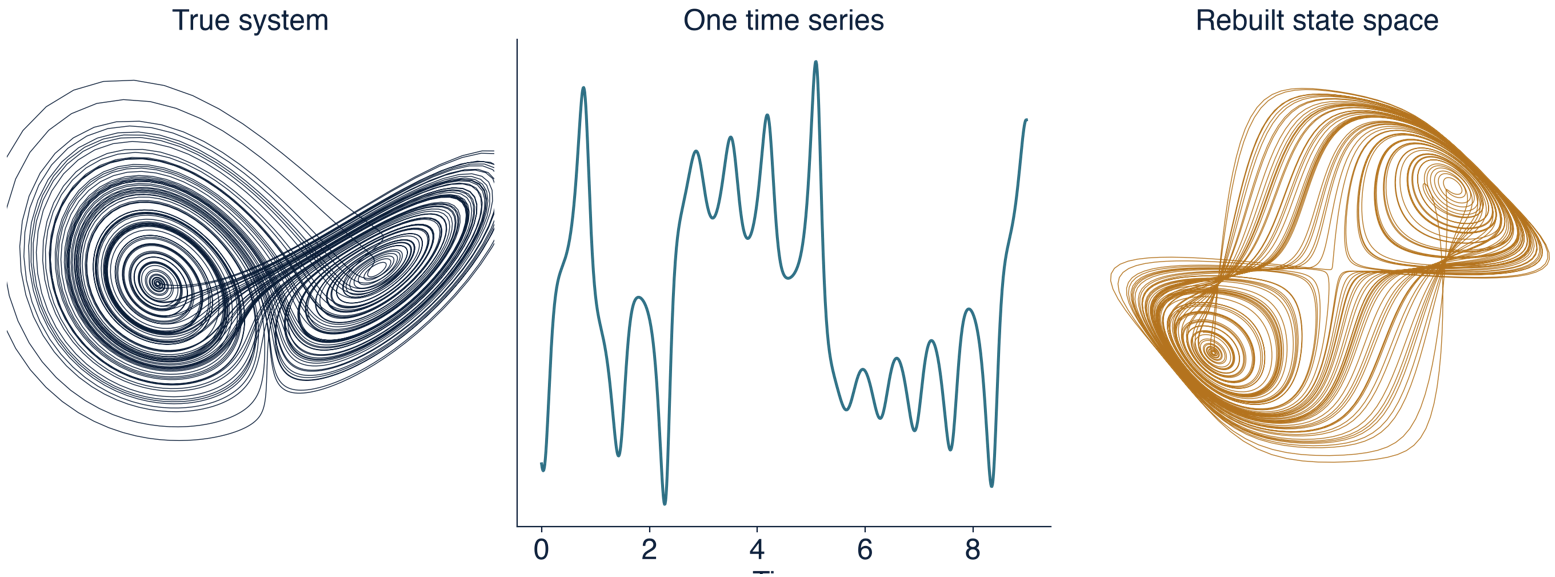
SeeWandel-Climate · Pomati lab and colleagues at Eawag
Luis J. Gilarranz · Stefanie Eyring · Stephan B. Munch · Tanya L. Rogers · Francesco Pomati
Data: IGKB, IBKF, University of Konstanz (D. Straile)

Supplementary material

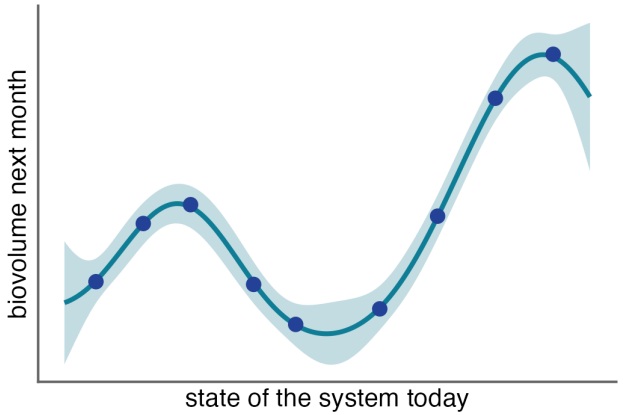
Methods in detail



What is GP-EDM?



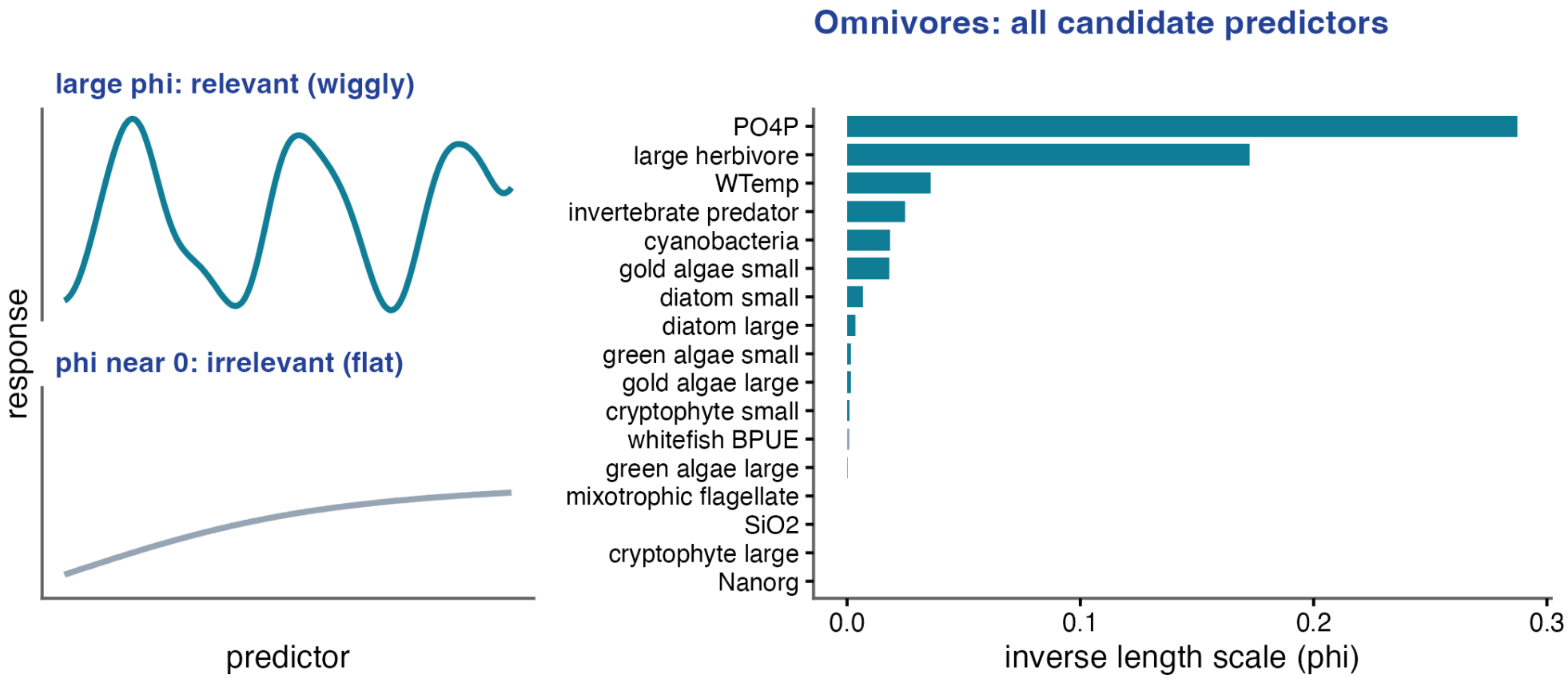
Gaussian process: no fixed equation



Next month = f(past states, drivers)

f learned from the data: Gaussian process

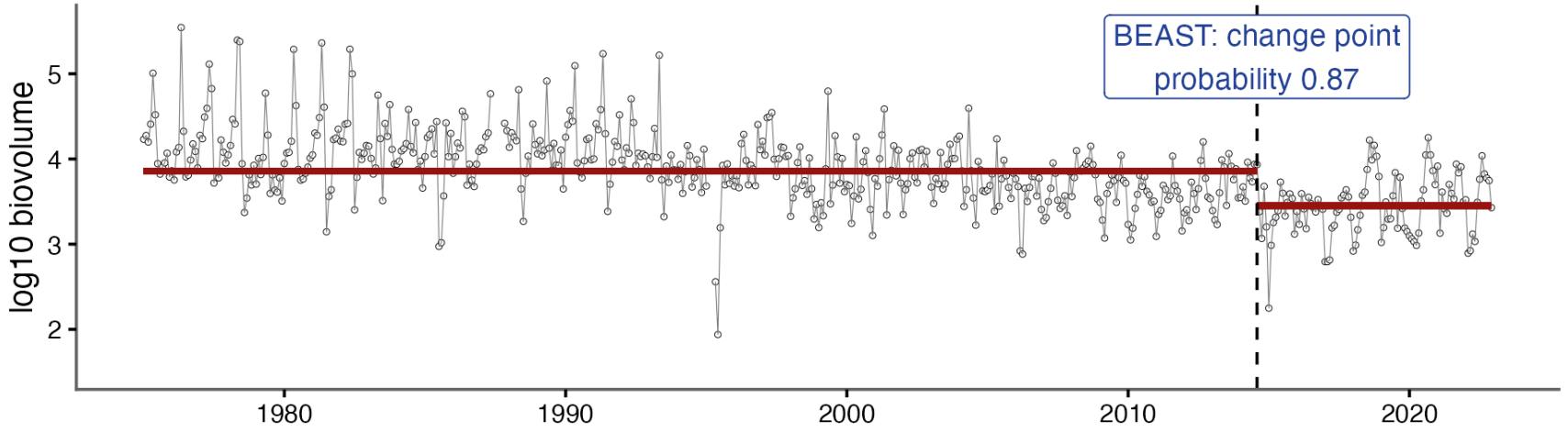
Automatic relevance determination (ARD)



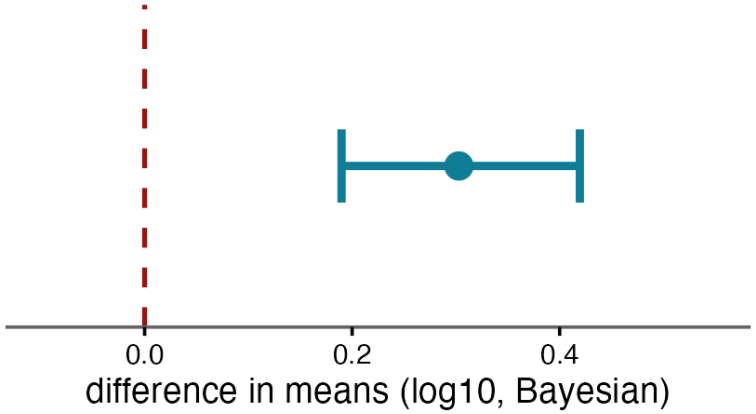
One length scale per predictor: **phi near 0** = predictor dropped

Detecting regime shifts: BEAST and two criteria

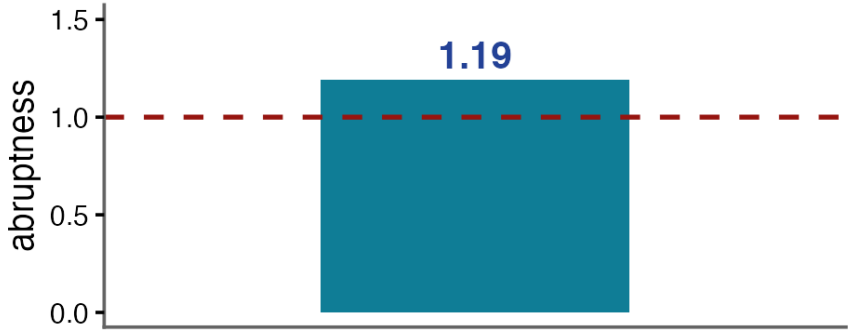
1: Candidate: BEAST change point



2: 95 % interval excludes 0



3: Abruptness > 1



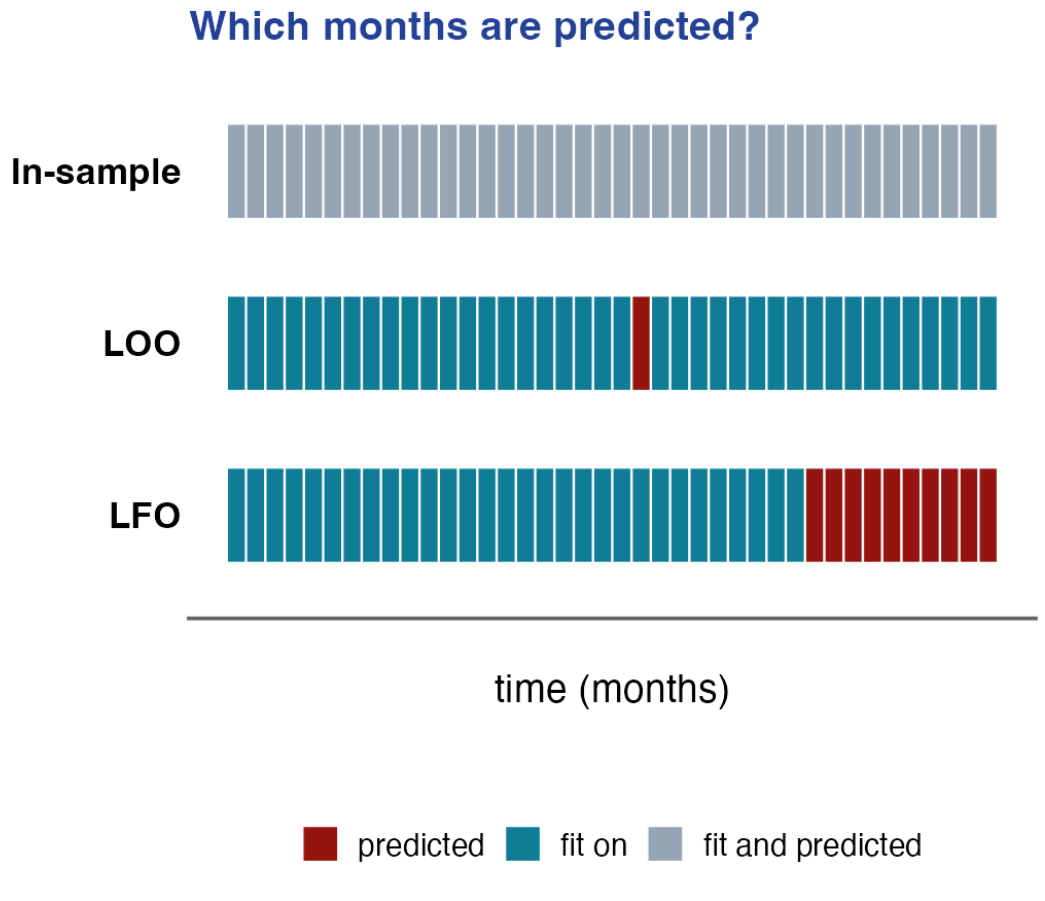
Kept if **both** hold

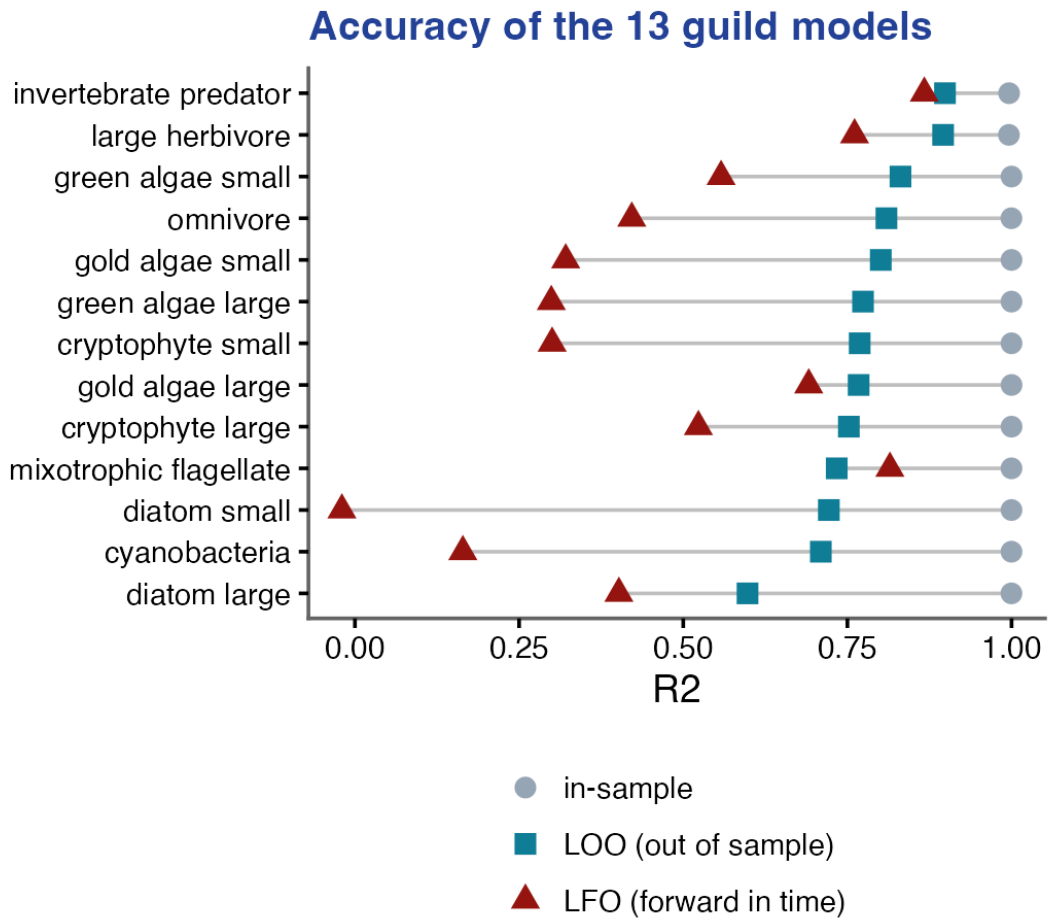
2 interval excludes 0

3 abruptness > 1

abruptness = |difference| / mean SD of the two periods

In-sample versus out-of-sample R^2





In-sample: $R^2 = 1$, a GP interpolates

LOO: one month left out

LFO: forecast the last 10 %

References

Eyring S, Merz E, Reyes M, Ntetsika P, et al. 2025. Distinct phytoplankton size classes respond differently to biotic and abiotic factors. ISME Communications 5: ycae148.

Gilarranz LJ, Narwani A, Odermatt D, Siber R, Dakos V. 2022. Regime shifts, trends, and variability of lake productivity at a global scale. PNAS 119: e2116413119.

Medeiros LP, Sorenson DK, Johnson BJ, Palkovacs E, Munch SB. 2025. Revealing unseen dynamical regimes of ecosystems from population time-series data. PNAS 122: e2416637122.

Merz E, Saberski E, Gilarranz LJ, Isles PDF, et al. 2023. Disruption of ecological networks in lakes by climate change and nutrient fluctuations. Nature Climate Change 13: 389–396.

Munch SB, Poynor V, Arriaza JL. 2017. Circumventing structural uncertainty: a Bayesian perspective on nonlinear forecasting for ecology. Ecological Complexity 32: 134–143.

Ntetsika P, Eyring S, Merz E, Reyes M, et al. 2026. Biotic interactions shape the realized niche of toxic cyanobacteria. Nature Ecology & Evolution 10: 1434–1448.

Takens F. 1981. Detecting strange attractors in turbulence. Lecture Notes in Mathematics 898: 366–381.

To S-W, Chakraborty S, Acevedo-Trejos E, Pomati F, et al. 2026. Future warming reshapes inorganic nutrient and plankton dynamics in a temperate lake. Limnology and Oceanography 71: e70447.

Figure of Lake Constance (map, phosphorus, temperature): D. Straile, University of Konstanz. Data: IGKB, IBKF.